

Chapter 31. Mechanical Ventilation in Chronic Obstructive Pulmonary Disease

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Mechanical Ventilation in Chronic Obstructive Pulmonary Disease: Introduction

Acute exacerbation of chronic obstructive pulmonary disease (COPD) is defined as an increase in dyspnea, cough, or sputum production that requires therapy. The annual rate of exacerbations is 0.5 to 3.5 per patient.¹ Hospitalization rates range from 0.1 to 2.4 per patient per year.¹ Most exacerbations are caused by viral infections, such as *Rhinovirus* or influenza species or bacterial infections, such as *Haemophilus influenzae*, *Streptococcus pneumoniae*, *Moraxella catarrhalis*, or *Pseudomonas* species.² Occasionally, exacerbations are caused by air pollution and other environmental factors.² When evaluating a patient suspected of having an exacerbation, concurrent conditions such as pulmonary emboli, pneumothorax, and congestive heart failure should be clinically excluded.² In about one-third of cases, no underlying etiology is identified.³

The clinical presentation of exacerbations of COPD is highly variable. Most patients require only an increase of maintenance medications, while others develop frank respiratory failure and require ventilator assistance.^{4,5} The goals of ventilator assistance are to decrease respiratory distress and dynamic hyperinflation, to improve gas exchange, and to buy time for resolution of the processes that triggered the episode of acute respiratory failure.

This chapter focuses on the aspects of ventilator management that are unique for patients with COPD. General principles of ventilator management are covered in other chapters—such as ventilator modes, bronchodilator therapy, weaning, and so on—and only the aspects specific to COPD are discussed in the present chapter. More than is the case with any other group of patients, clinical decision making and ventilator management in COPD is predicated on a detailed knowledge of the underlying pathophysiology.

Pathophysiologic Features Relevant to Ventilator Support

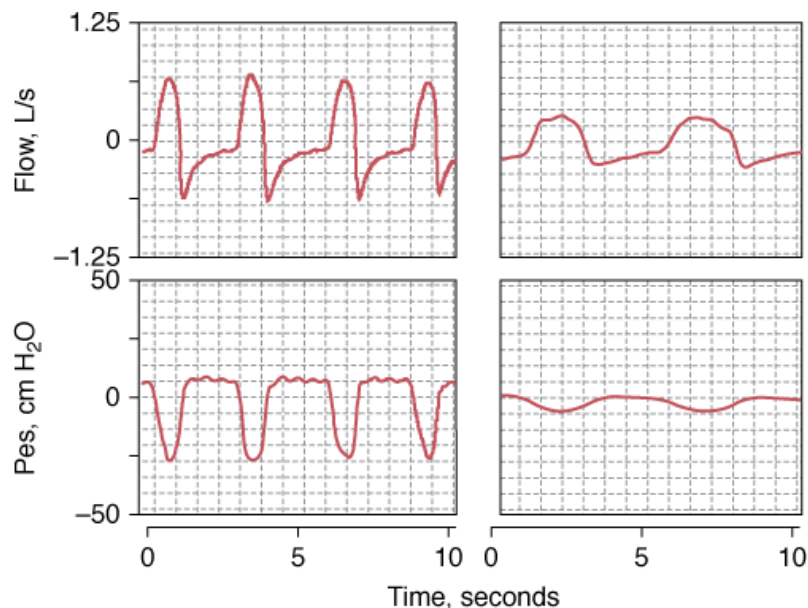
The basic physiologic abnormalities of patients who experience acute respiratory failure in COPD include deteriorations in respiratory mechanics, respiratory muscle function, and gas exchange.

Deterioration of Respiratory Mechanics

Increased Inspiratory Airway Resistance

In stable patients with COPD, inspiratory flow resistance is approximately 6 cm H₂O/L/s above normal.⁶ During an episode of acute respiratory failure, resistance can increase by an additional 6 cm H₂O/L/s or more.^{7,8} Mechanisms responsible for the increase include bronchospasm, airway inflammation, and mucus production.⁹ Increases in inspiratory resistance and respiratory motor output lead to increased inspiratory effort (Fig. 31-1).⁷ In patients with severe COPD, the pressure output of the inspiratory muscles during resting breathing is three times higher than in healthy subjects: average pressure-time products of 259 to 341 versus 94 cm H₂O • second per minute.^{8,10} Values are five times higher when patients are in respiratory distress.^{8,11}

Figure 31-1



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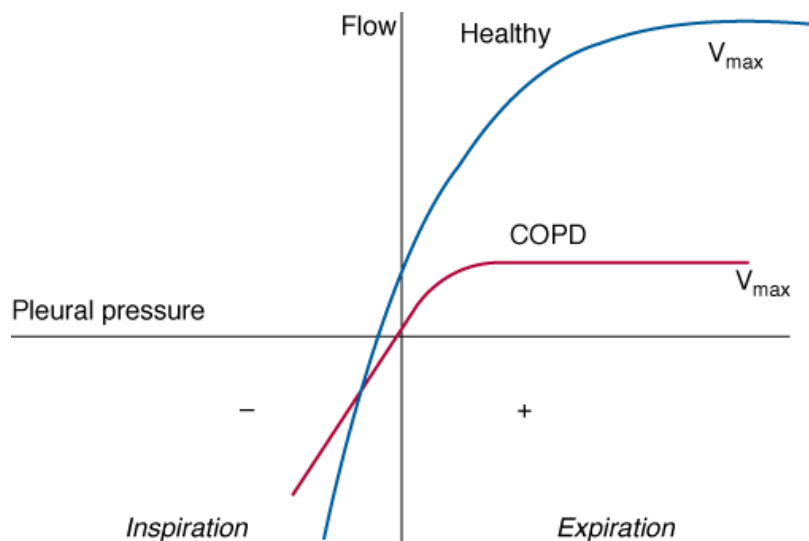
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Respiratory effort during spontaneous respiration. Recordings of flow (*inspiration upward*) and esophageal pressure (*Pes*) in a patient with chronic obstructive pulmonary disease (COPD) in respiratory failure (*left panel*) and in a stable patient with COPD (*right panel*). The patient in respiratory failure exhibits a steeper fall and greater excursion in esophageal pressure than does the stable patient, signifying greater respiratory motor output. Despite the fivefold larger excursion in *Pes* in the patient in respiratory failure, peak inspiratory flow is only two times more than in the stable patient, signifying more abnormal mechanics in the former than in the latter patient. The patient in respiratory failure exhibits tachypnea, reflected by the shorter respiratory cycle, and exhibits a supramaximal flow transient at the beginning of exhalation, which is typical of expiratory flow limitation.

Increased Expiratory Airway Resistance

In healthy subjects, inspiratory and expiratory airway resistance are of similar magnitude.^{6,12} In contrast, patients with COPD have greatly increased expiratory resistance.¹²⁻¹⁴ In ventilated patients with COPD, the ratio of expiratory to inspiratory flow resistance ranges from 3.8 at low lung volumes to 1.6 at high lung volumes.¹⁴ The increased resistance at low lung volumes is related to dynamic narrowing of the small airways during exhalation.¹⁵ This narrowing is thought to result from damage to the elastic scaffold surrounding the airways¹⁶ and as well as to the “wave speed limitation” of the expiratory flow—the tracheobronchial tree cannot adjust an airflow more rapidly than the velocity at which pressure travels along the airways.¹⁵ Small-airway narrowing and wave-speed limitation reduce maximum flow during exhalation with the development of expiratory flow limitation (Fig. 31-2). Expiratory flow limitation occurs in 60% of stable patients during resting breathing,¹⁷ and universally during episodes of acute respiratory failure.¹⁸

Figure 31-2



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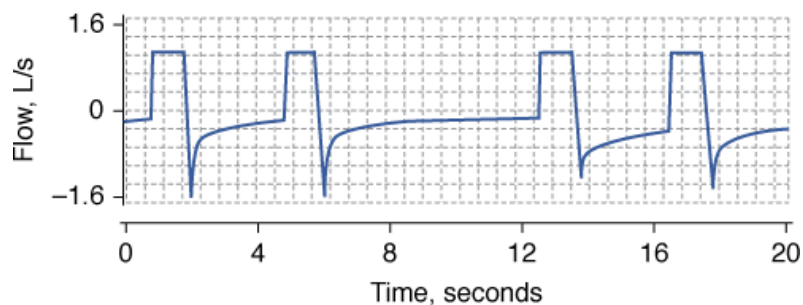
Schematic of the isovolume pressure–flow relationship. Patients with chronic obstructive pulmonary disease (COPD) exhibit an initial diagonal segment, where increases in pressure produce increases in airflow (the effort-dependent region on the *left*), followed by a flat portion, where increases in pressure do not produce increases in flow (the effort-independent region on the *right*). Compared with a healthy subject, the slope of the initial diagonal segment is decreased, indicating an increase in airway resistance, and maximum flow is much reduced in the remaining portion secondary to expiratory flow limitation. (Used, with permission, from Tobin et al.¹⁶)

Patients with expiratory flow limitation experience air trapping whereby activation of the expiratory muscles increases alveolar pressure without decreasing the end-expiratory lung volume below the relaxation volume of the respiratory system (V_{rel}).¹⁵ The inability to decrease end-expiratory lung volume below V_{rel} has several negative consequences. First, expiratory muscle contraction—a constrained response of the respiratory centers to increased ventilatory demands—does not achieve storage of elastic energy in the respiratory system at end exhalation.⁴ Second, relaxation of the expiratory muscles at the onset of inhalation cannot assist the inspiratory muscles in expanding the respiratory system during inhalation. Third, the entire burden of breathing is borne by the inspiratory muscles. Fourth, recruitment of the expiratory muscles dissipates precious energy substrates. That is, expiratory muscle recruitment in COPD is harmful,¹⁹ and it can account for 66% of the variation in Borg scale ratings of difficulty in breathing.²⁰

Dynamic Hyperinflation and Intrinsic Positive End-Expiratory Pressure

The minimum time required to exhale from a given lung volume to V_{rel} is determined by the maximum expiratory flow.¹⁵ When the time for exhalation (T_E) is less than this minimum time, inhalation will begin before the respiratory system has returned to V_{rel} —a state known as dynamic hyperinflation (Fig. 31-3).¹⁶ Dynamic hyperinflation (a “volume” phenomenon) is almost invariably associated with an increase in end-expiratory elastic recoil of the respiratory system (a “pressure” phenomenon).¹⁵ This increase in end-expiratory elastic recoil has been called auto-positive end-expiratory pressure (auto-PEEP)²¹ or intrinsic PEEP (PEEPi).²² Dynamic hyperinflation and PEEPi have been reported in all ventilated patients with COPD.²³ Increases in dynamic hyperinflation during exacerbations are proportionally greater than increases in airflow obstruction.²⁴

Figure 31-3



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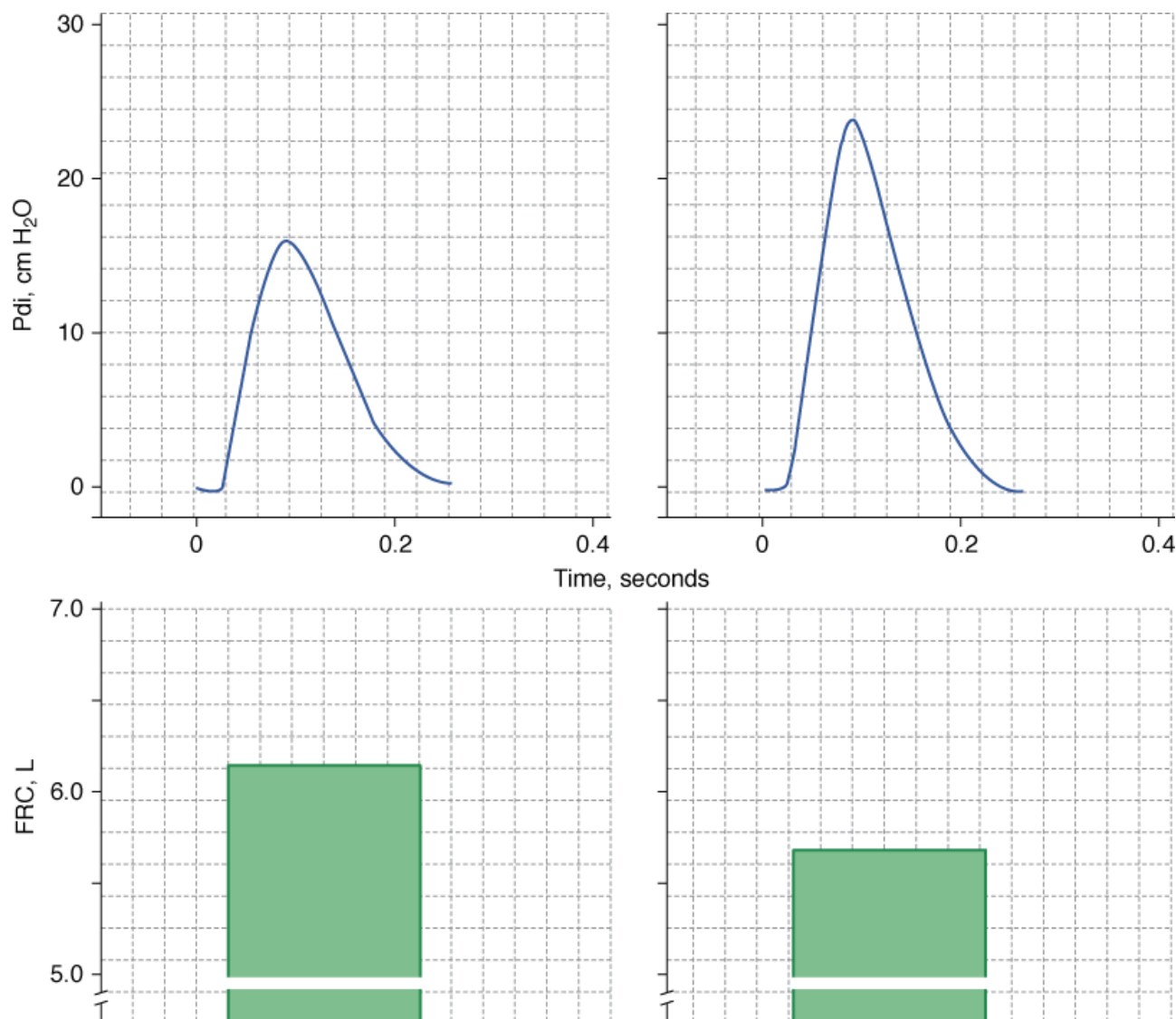
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Persistent expiratory flow at end-exhalation. Recording of flow (*inspiration upward*) in a patient with chronic obstructive pulmonary disease (COPD) and respiratory failure receiving controlled mechanical ventilation. Whether the duration of exhalation is 3 seconds (first and third breaths) or 6 seconds (second breath), expiratory flow at end-exhalation is always present. Persistent expiratory flow at end-exhalation signifies that inhalation begins before the respiratory system has returned to its relaxation volume—dynamic hyperinflation.

Dynamic hyperinflation and PEEPi can fluctuate widely as a result of several factors.¹⁵ First, bronchospasm, mucosal edema, and sputum inspissation can abruptly worsen expiratory flow limitation. Second, patients are often tachypneic and, thus, T_E is shortened.²⁵ Finally, ventilatory demands can fluctuate because of sudden episodes of anxiety or deterioration of gas exchange.¹⁵

Acute worsening in dynamic hyperinflation is beneficial because it optimizes expiratory flow by limiting dynamic airway narrowing during exhalation²⁶ and by increasing the elastic recoil of the lung and, less so, of the chest wall.¹⁸ It is detrimental because it decreases the effectiveness of the respiratory system through several mechanisms. First, at the start of inhalation patients have to first generate negative inspiratory pressure equal in magnitude to the value of PEEPi before inspiratory flow can be initiated (threshold inspiratory load).^{8,27} In patients with COPD recovering from an episode of respiratory failure necessitating mechanical ventilation, Jubran and Tobin⁸ reported that, during a spontaneous breathing trial, approximately 20% to 25% of the inspiratory muscle effort was required to overcome PEEPi. Second, dynamic hyperinflation impairs respiratory muscle function by worsening the length-tension relationship of the muscles (Fig. 31-4), by decreasing the zone of apposition,²⁸ and by interfering with inspiratory muscle perfusion.²⁹ Finally, dynamic hyperinflation increases end-inspiratory lung volume.¹⁸ When excessive, increases in end-inspiratory lung volume can reduce lung compliance, increase the elastic work of breathing, and can cause alveolar overdistension. Alveolar overdistension can lead to hemodynamic compromise through several mechanisms. First, it can reduce right-ventricular preload as a result of impaired venous return.²¹ Second, it can raise right-ventricular afterload by increasing pulmonary vascular resistance.²¹ The increased right-ventricular afterload can increase the right ventricular end-diastolic volume and shift the interventricular septum toward the left ventricle. This shift can impair left-ventricular filling (ventricular interdependence). Finally, the threshold load imposed on the inspiratory muscles can increase left-ventricular afterload secondary to increased negative intrapleural pressure during inhalation.²¹ All these phenomena can severely decrease cardiac output, effects exaggerated in COPD because the abnormally compliant lungs transmit a high fraction of alveolar pressure to intrathoracic vessels.³⁰

Figure 31-4



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Twitch transdiaphragmatic pressure (P_{di}) elicited by phrenic nerve stimulation (*upper panels*) and functional residual capacity (FRC ; *lower panels*) in a patient with chronic obstructive pulmonary disease (COPD) before (*left*) and after (*right*) lung volume reduction surgery. The higher twitch P_{di} after surgery was in part caused by decrease in operating lung volume as demonstrated by decrease in FRC . (Data from Laghi et al.¹⁰)

Deterioration of Respiratory Muscle Function

Respiratory Muscle Weakness

Patients with COPD do not generate as much negative maximal inspiratory pressures as do healthy subjects.⁴ This decrease in respiratory muscle strength may result from several factors, including greater protein degradation of muscle fibers,^{31,32} malnutrition, sepsis, systemic steroids, and ventilator mode.^{4,28} In some patients, however, the inspiratory weakness can be completely explained by hyperinflation-induced muscle shortening.³³ In addition, diaphragmatic myofibers of patients with COPD are also more susceptible to sarcomere disruption when subjected to an acute inspiratory load than are the diaphragms of healthy controls.³⁴

Patients with COPD have an increased risk of coronary artery disease, and 20% to 30% have chronic heart failure.³⁵ Increased stress on the myocardium during acute respiratory failure could overwhelm an already impaired cardiac reserve.^{36,37} An increased load resulting

from interstitial edema secondary to acute left-ventricular failure together with decreased blood flow to the respiratory muscles may markedly impair respiratory muscle performance.³⁸

Respiratory Muscle Fatigue

Contractile fatigue occurs when a sufficiently large respiratory load is applied over a sufficiently long period of time. Contractile fatigue can be brief or prolonged.²⁸ Short-lasting fatigue results from accumulation of inorganic phosphate, failure of the membrane electrical potential to propagate beyond T-tubules, and to a much lesser extent intramuscular acidosis.²⁸ In nine patients who developed acute respiratory failure during a weaning trial (four of whom had COPD), Brochard et al³⁹ reported electromyographic signs suggestive of incipient short-lasting diaphragmatic fatigue. Short-lasting fatigue appears to have a protective function, because it can prevent injury to the sarcolemma caused by forceful muscle contractions.⁴⁰ Long-lasting fatigue⁴¹ is consistent with the development of, and recovery from, muscle injury.^{28,40}

To assess whether patients in acute respiratory failure develop long-lasting fatigue, Laghi et al¹¹ measured the contractile response of the diaphragm to phrenic nerve stimulation in sixteen patients being weaned from mechanical ventilation. In that study, no patient developed long-lasting fatigue of the diaphragm. The investigators concluded that patients displayed clinical manifestations of severe respiratory distress for a substantial time before they would develop fatigue.

Deterioration of Gas Exchange

Abnormal distribution of ventilation–perfusion ($\dot{V}_A/\dot{Q}$) ratios⁴² and decreased mixed venous oxygen tension are common causes of hypoxemia during exacerbations.^{43,44} In patients without alveolar pathologies (pneumonia, pulmonary edema) shunt is almost negligible.⁴³

Hypercapnia is present in nearly half of patients admitted to a hospital with an acute exacerbation of COPD.⁴⁵ Hypercapnia plus respiratory acidosis is almost universal in patients with COPD requiring mechanical ventilation.⁴⁵

Excluding patients who are on the verge of respiratory arrest, most patients with COPD in respiratory failure who develop hypercapnia experience a decrease in alveolar ventilation ($\dot{V}_A$) as a result of a change in the pattern of breathing and not as a result of decreased respiratory drive.²⁵ The dominant finding of this pattern of breathing is a shortening of inspiratory time (T_I) and T_E .²⁵ These combined changes lead to a marked increase in respiratory frequency and decrease in tidal volume (V_T)—rapid shallow breathing. The decrease in V_T results in increased dead-space-to-tidal-volume ratio (V_D/V_T) or dead space ventilation that is not compensated by the increase in minute ventilation. In a study of seventeen patients with COPD requiring mechanical ventilation, Tobin et al²⁵ reported that the combined changes in V_T and respiratory frequency accounted for 81% of increase in partial pressure of arterial carbon dioxide (P_{aCO_2}) in the ten patients who developed respiratory distress when disconnected from mechanical ventilation. Whether rapid shallow breathing is an adaptive response to the mechanical constraints of acute dynamic hyperinflation that minimizes dyspnea and lessens the risk of respiratory muscle fatigue remains to be determined.

In some patients with COPD, hypercapnia can be worsened by the administration of supplemental oxygen.⁴⁵ This risk is significant when the inspired oxygen concentration exceeds 30%. Mechanisms that may contribute to CO_2 retention include a decrease in hypoxic ventilatory response consequent to the administration of oxygen; an increase in dead space consequent to release of hypoxic vasoconstriction and, thus, worsening of $\dot{V}_A/\dot{Q}$ relationships; and the Haldane effect (for any given amount of CO_2 bound to hemoglobin P_{aCO_2} is considerably higher in the presence of a high versus a low oxygen saturation).¹⁶

Ventilator Assistance

The many pathophysiologic defects considered above have direct bearing to the judicious administration of mechanical assistance in patients with COPD in respiratory failure. Physicians considering the need to implement mechanical ventilation in these patients have to

answer five basic questions:

1. When should mechanical ventilation be started, and which goals should be pursued?
2. How should mechanical ventilation be delivered—noninvasive versus invasive?
3. What setting and mode of mechanical ventilation should be used?
4. When—and how—should mechanical ventilation be discontinued?
5. What can be done when weaning is difficult or impossible?

Ventilator Assistance: Indications

Most patients with mild exacerbations of COPD (70% to 90% of those hospitalized for an exacerbations)^{5,45,46} can be cared for with oxygen and pharmacotherapy alone. The remaining 10% to 30% require ventilator assistance.^{5,45–47}

Indications for Noninvasive Positive-Pressure Ventilation

Noninvasive positive-pressure ventilation (NIPPV) has become the first line intervention for most patients with COPD in acute respiratory failure (Table 31-1).⁵ NIPPV decreases the need for endotracheal intubation and the rate of ventilator-related complications, intensive care unit (ICU) admissions, and in-hospital and 1-year mortality.^{47–49}

Table 31-1: Indications and Contraindications for Noninvasive Ventilation in Exacerbations of Chronic Obstructive Pulmonary Disease**Indications***Clinical observations*

Moderate to severe dyspnea

Tachypnea

Accessory muscle use, abdominal paradox

*Impaired gas exchange*Acute or acute-on-chronic hypercapnic respiratory failure ($Pa_{CO_2} > 45$ mm Hg, pH < 7.35)Hypoxemia ($Pa_{O_2}/F_{I_{O_2}} \leq 200$)^a**Contraindications***Absolute*

Bradypnea, respiratory arrest, immediate need for intubation

Life-threatening hypoxemia

Unable to fit mask

Upper airway obstruction

Undrained pneumothorax

Vomiting/severe upper gastrointestinal bleeding

Relative

Agitated, uncooperative

Severe hypercapnic encephalopathy (Glasgow Coma Scale score < 10)

Inability to protect the airway

Impaired swallowing or cough

Excessive secretions

Recent upper airway or upper gastrointestinal surgery

Multiorgan failure

Medically unstable

Uncontrolled cardiac ischemia or arrhythmia

Hypotensive shock

Abbreviations: $F_{I_{O_2}}$, fractional inspired oxygen concentration; Pa_{CO_2} , partial pressure of arterial carbon dioxide; Pa_{O_2} , partial pressure of arterial oxygen.

^aNoninvasive ventilation should be used with caution in exacerbations of chronic obstructive pulmonary disease (COPD) accompanied with hypoxemia.

In most studies, the rate of failure with NIPPV is inversely related to the severity of respiratory acidosis. When pH is 7.30 to 7.34 on admission, the rate of NIPPV failure is 10% to 20%;^{47,50} when pH is 7.25 to 7.30, NIPPV failure is 30% to 40%;^{51,52} and when pH is less than 7.25, failure is 50% to 60%.^{53,54} These rates, however, have to be interpreted with caution. In some series, NIPPV failure in patients with very low pH (7.13 to 7.20) on admission was only 14% to 35%.^{55–57}

Failure with NIPPV usually occurs in the first hours of ventilator support.⁵⁸ Mask intolerance, uncontrollable leaks, and lack of improvement in gas exchange are the most common causes of failure.^{59,60} Early improvement should not lull the physician into a false sense of security: Approximately 20% of initial responders (first 48 hours) experience a second episode of acute respiratory failure.⁶⁰ Late-onset deterioration is more likely in patients with worse clinical condition before admission, and more complications at admission.⁶⁰ Patients who experience late failure have a poor in-hospital prognosis, particularly if they are not promptly intubated.⁶⁰

Indications for Noninvasive Negative Pressure Ventilation

Negative pressure ventilation (iron lung ventilation) has been successfully used in patients with COPD and severe respiratory acidosis ($\text{pH} < 7.25$).^{61–63} (See [Chapter 16](#).)

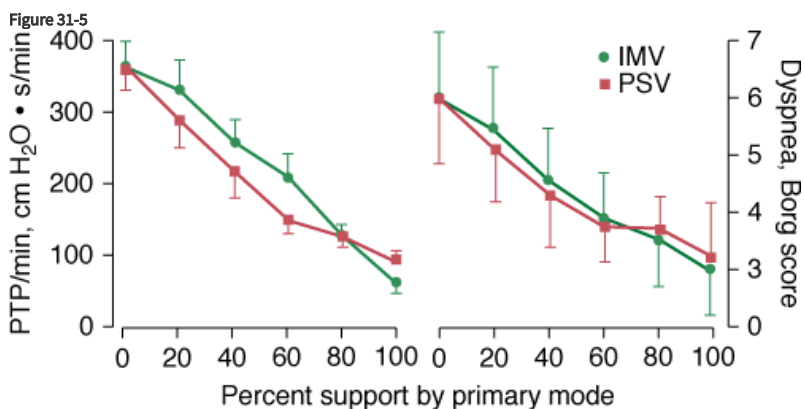
Indications for Invasive Ventilation

NIPPV and negative pressure ventilation do not protect the upper airway in patients who are gasping, have respiratory arrest or have lost consciousness.⁴⁷ NIPPV and negative pressure ventilation are also unsafe and poorly tolerated in patients requiring high levels of inspiratory airway pressure (>25 to $30 \text{ cm H}_2\text{O}$) and inspired oxygen ($\text{FI}_{\text{O}_2} > 0.60$). These patients, as well as those who are hemodynamically unstable despite fluid resuscitation and vasopressors, and those who are agitated, uncooperative, or who fail noninvasive ventilation, require endotracheal intubation and invasive ventilation (see [Table 31-1](#)). Despite earlier concerns, severe hypercapnic encephalopathy (Glasgow Coma Scale score ≤ 8 to 10) should not—a priori—be considered an absolute contraindication to NIPPV.^{55,57,64,65} If NIPPV is used in encephalopathic patients, opinions vary as to the advisability of inserting a nasogastric tube to decrease the risk of aspiration of gastric contents.^{55,57,64}

Goals of Ventilator Assistance

The primary goals of ventilator assistance are to decrease inspiratory effort and respiratory distress, to minimize dynamic hyperinflation and PEEPi, to lessen respiratory acidosis, and to improve hypoxemia.

Most patients with COPD in respiratory failure are dyspneic and have clinical signs of increased work of breathing: nasal flaring, vigorous activity of the sternomastoid muscles, tracheal tug, recession of the suprasternal, supraclavicular and intercostal spaces, paradoxical motion of the abdomen, and pulsus paradoxus.⁶⁶ In these patients, invasive^{67,68} and noninvasive^{69–71} ventilation can decrease inspiratory effort and respiratory distress ([Fig. 31-5](#)).



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Inspiratory effort, quantitated as by esophageal pressure-time product per minute (PTP/min), and dyspnea, quantitated by Borg score in eleven ventilator-dependent patients with chronic obstructive pulmonary disease (COPD) while receiving pressure-support ventilation (PSV) and intermittent mandatory ventilation (IMV). *Left panel*: PTP/min decreased as the level of PSV or IMV was increased. At proportional levels of ventilatory assistance, PTP/min was not different during PSV and IMV ($p < 0.0005$ in both instances). *Right panel*: Dyspnea decreased with increasing levels of PSV or IMV ($p < 0.05$ in both instances). Bars represent $\pm$ standard error (SE). (Modified, with permission, from Leung et al.⁶⁷)

Mechanical ventilation can both worsen or improve dynamic hyperinflation and PEEPi.¹⁵ Dynamic hyperinflation and PEEPi should be suspected in any patient with persistent expiratory flow (see [Fig. 31-3](#)) at end-exhalation.¹⁵ The magnitude of end-expiratory flow, however, bears little relation to the magnitude of PEEPi.³⁰

Mechanical ventilation improves hypercapnia by decreasing CO_2 production and, more importantly, by increasing $\dot{V}_A$. Vigorous increases in minute ventilation should be avoided, because of risk of iatrogenic dynamic hyperinflation. When dynamic hyperinflation is a concern,

and provided that intracranial hypertension and overt hemodynamic instability do not exist, acceptance of acidemia ($\text{pH} > 7.2$) may be reasonable.⁷²

In addition to worsening dynamic hyperinflation, overzealous ventilation can cause life-threatening alkalosis.⁶⁶ Severe alkalosis can cause coronary artery spasm, central nervous system hypoperfusion, myoclonus, asterixis, and seizures. Other effects of alkalosis include increase in hemoglobin affinity for oxygen, and, in the presence of increased shunt, a possible worsening of the $\dot{V}_A/\dot{Q}$ relationship (secondary to a decrease in hypoxic pulmonary vasoconstriction).⁶⁶ Excessive ventilation, over time, causes bicarbonate wasting by the kidney. In patients who retain CO_2 when clinically stable, this renal wasting of bicarbonate will increase ventilatory demands during weaning.

Choice of Ventilator Mode

A wide variety of ventilator modes are used in COPD. These include pressure-support ventilation (PSV), assist-control ventilation (ACV) that can be either volume-cycled or pressure-cycled (PCV), intermittent mandatory ventilation, and, more recently, proportional-assist ventilation and neurally adjusted ventilatory assist. Each of these modes is covered in detail in other chapters.

It is unknown whether one ventilator mode is superior to another in patients with COPD. I summarize here my own preferences, and the physiologic rationale behind my choices. In patients requiring NIPPV, I usually use PSV^{5,73} based on two considerations. First, unlike volume-cycled modes (ACV, intermittent mandatory ventilation), PSV allows breath-by-breath titration of support. The hope is to improve patient comfort and thus achieve greater success with NIPPV. Second, PSV is equally as effective as volume-cycled modes in decreasing inspiratory effort (see Fig. 31-5). I add a moderate amount of external PEEP to counterbalance PEEPi—a combination that lowers inspiratory work more than the use of PSV or PEEP alone.^{74,75} Persistent respiratory distress despite high levels of support can result from expiratory muscle recruitment.⁷⁶ This recruitment can be caused by inadequate “cycling-off” criteria of PSV: In patients with COPD the decay in inspiratory flow during PSV is less steep than in patients with normal respiratory mechanics.^{76,77} When expiratory muscle recruitment does occur an increase in the threshold of peak inspiratory flow to cycle off PSV can improve patient-ventilator interaction.^{77,78}

Persistent respiratory distress despite high levels of support can result from excessive load on the respiratory muscles. In this situation, an increase in inspiratory pressurization rate—the time needed to reach the target inspiratory pressure—can decrease inspiratory effort.⁷⁹ Unfortunately, increases in pressurization rate during NIPPV are associated with increased air leaks and decreases a patient’s tolerance of NIPPV.⁷⁹ When respiratory distress persists despite high levels of support, it may be more prudent to intubate a patient.

In intubated patients, I mostly use ACV⁸⁰ and, occasionally, PCV. I prefer ACV and PCV over PSV in intubated patients because these patients commonly receive sedation. Sedation can decrease respiratory motor output and thus promote alveolar hypoventilation. When using ACV I usually set the inspiratory flow waveform in the square pattern to facilitate monitoring of mechanics. When intubated patients with COPD display a high drive, I may use PCV. With this mode, patients have more control over the peak flow than with ACV. I avoid controlled mechanical ventilation because it is associated with the early development of respiratory muscle atrophy and damage.^{81,82}

Proportional-assist ventilation and neurally adjusted ventilatory assist are two novel modes, but it is not known whether they will prove superior to traditional ventilator modes.

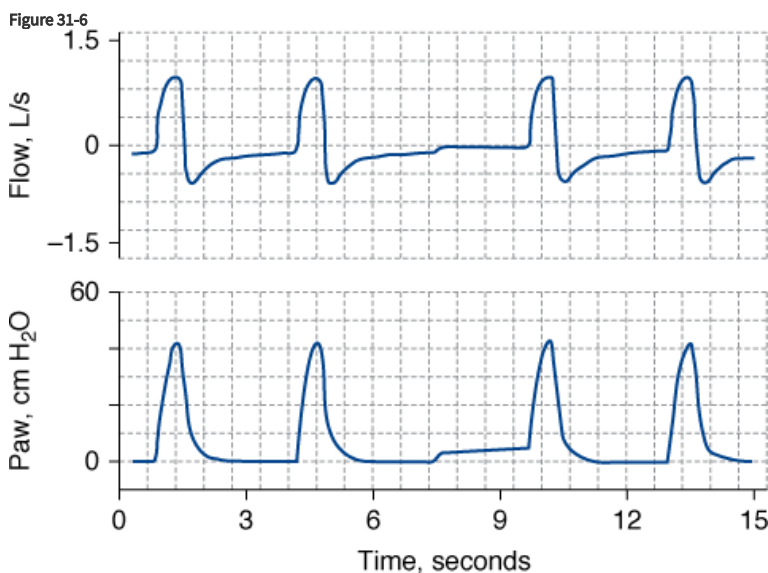
Intrinsic and External Positive End-Expiratory Pressure

The most valuable approach to reduce or eliminate PEEPi is to diminish minute ventilation,³⁰ even if this means development of alveolar hypoventilation—permissive hypercapnia.⁸³ In patients with PEEPi ventilated with volume-cycled modes, clinicians often increase inspiratory flow to decrease T_I . This is done hoping to prolong T_E . Increases in inspiratory flow, however, commonly lead to an increase in respiratory rate^{84,85} and the anticipated reduction in T_E could increase PEEPi. Laghi et al⁸⁵ studied this phenomenon in ten patients with COPD while they were receiving ACV with a backup rate of 1 breath/min. As expected, an increase in flow from 30 to 90 L/min increased the respiratory rate from 16 ± 1 to 21 ± 2 breaths/min. Despite the rise in rate, PEEPi decreased from 7 ± 1 to 6 ± 1 cm H_2O . The decrease in

PEEPi was the result of a paradoxical increase in T_E , which permitted extra time for lung deflation. The investigators reasoned that the shortened T_I secondary to the increase inspiratory flow combined with time-constant inhomogeneity of COPD caused overinflation of some lung units to persist into neural exhalation. Continued inflation during neural exhalation could stimulate the vagus, which prolongs expiratory time.^{84,85}

In patients with COPD and flow-limited physiology, low levels of external PEEP can maintain a patent (or nearly patent) channel between the central airway and the alveolar compartment at the start of inhalation. In these patients, application of low-level external PEEP reduces the effort to trigger the ventilator,^{14,86} improves patient-ventilator interaction,¹⁴ and reduces the effort to start inspiratory flow during weaning from mechanical ventilation.⁸⁷ External PEEP of 5 cm H₂O or less often suffices to reach these goals.⁸⁸ In patients with PEEPi, intravascular fluid expansion can increase blood pressure and cardiac output.²¹ Prompt disconnection from the ventilator can resolve hypotension secondary to PEEPi. If that is the case, mechanical ventilation should be restarted with lower minute ventilation.

The selection of external PEEP should be guided by measurement of static PEEPi (Fig. 31-6).¹⁵ When PEEPi is present, the rise in airway pressure during the occlusion maneuver should be maintained until a plateau is reached, usually in less than 4 seconds.¹⁸ The delayed plateau in the airway pressure signal is caused by stress adaptation phenomena and by time-constant inequalities.¹⁸ During controlled mechanical ventilation, application of a value of external PEEP at or above 50% of the static PEEPi improved ventilation-perfusion matching (Fig. 31-7)⁸⁹ and oxygenation without causing a decrease in cardiac output.⁹⁰



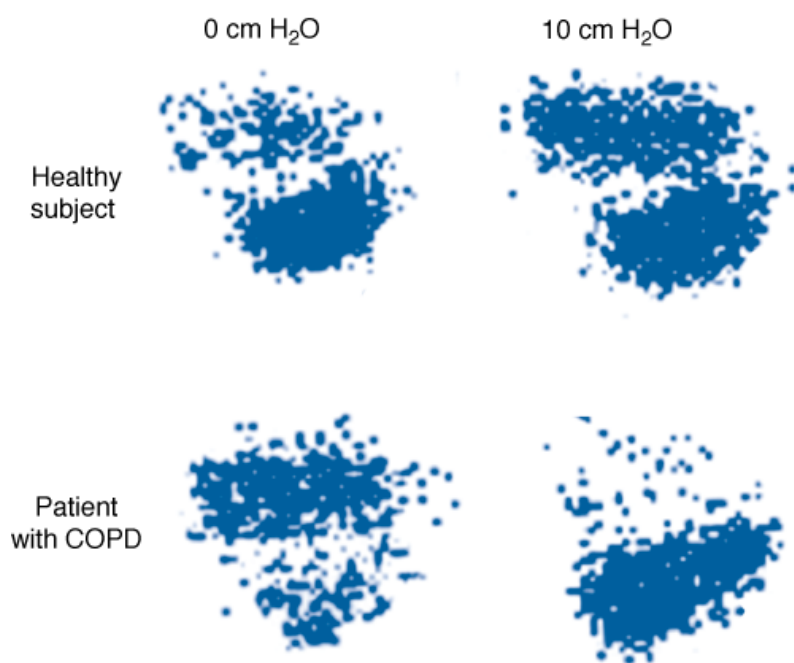
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Measurement of static intrinsic positive end-expiratory pressure (PEEPi) by single-breath end-expiratory airway occlusion in a patient with chronic obstructive pulmonary disease (COPD) during controlled mechanical ventilation. End-expiratory airway occlusion is carried out at the time when the third ventilator breath should have taken place. The end of occlusion to preocclusion difference in airway pressure (Paw) is the static PEEPi. In this example, static PEEPi is 6 cm H₂O.

Figure 31-7

External PEEP



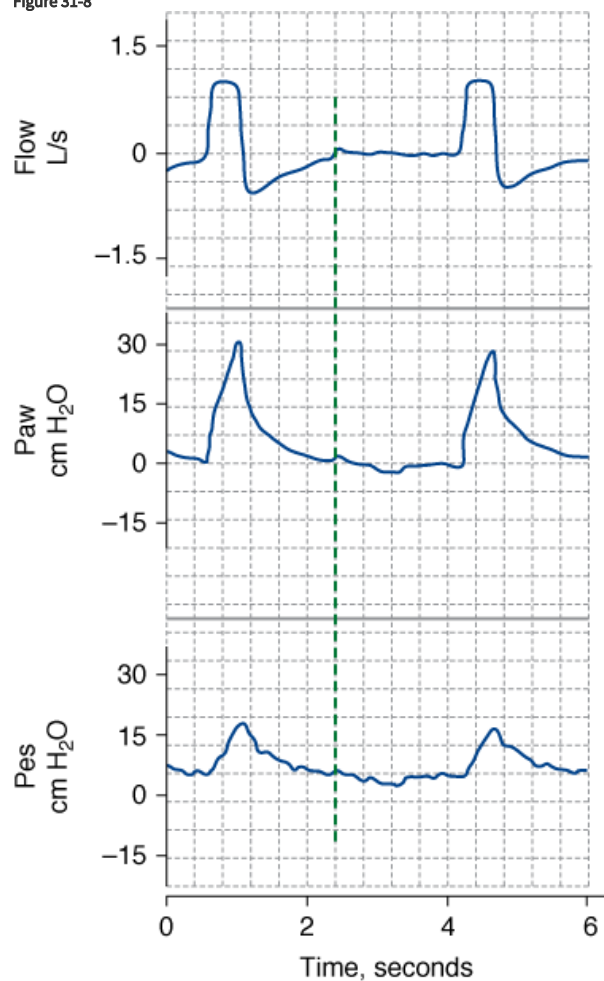
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Ventilation scans (radioactive krypton) in a healthy subject (*upper panels*) and in a patient with chronic obstructive pulmonary disease (COPD) (*lower panels*) resting in the left lateral decubitus position while receiving positive end-expiratory pressure (PEEP) of 0 cm H₂O (*left panels*) and 10 cm H₂O (*right panels*). At a PEEP of 0 cm H₂O, ventilation was distributed predominantly to the dependent regions in the healthy subject (*left upper panel*) and in the nondependent regions in the patient (*left lower panel*). PEEP 10 cm H₂O restored ventilation to the dependent regions in the patient probably because it prevented collapse of the dependent airways. (Modified, with permission, from Shim et al.⁸⁹)

When measuring static PEEPi with the occlusion method, it is impossible to detect any increased alveolar pressure distal to airways that are completely occluded during exhalation.⁹¹ In other words, the occlusion method estimates only the end-expiratory alveolar pressure that is measurable in the upper airway during tidal breathing.³⁰ Consequently, some experts advocate the measurement of end-inspiratory plateau airway pressure as external PEEP is titrated as a more sensitive tool to monitor the potential perils of dynamic hyperinflation.^{15,30,91}

The occlusion method is impractical and potentially inaccurate in patients who trigger most breaths during assisted ventilation (Fig. 31-8).¹⁵ These patients can be instrumented with an esophageal balloon catheter system to measure the deflection in esophageal pressure from the beginning of inspiratory effort to the onset of inspiratory flow (counterbalance method) (Fig. 31-9).¹⁵ This deflection is the so-called dynamic PEEPi. The counterbalance method is based on several assumptions.²⁷ First, the end-expiratory alveolar pressure represents the elastic recoil pressure of the relaxed respiratory system. Second, the change in esophageal pressure from the beginning of inspiratory effort to the onset of inspiratory flow reflects the inspiratory muscle pressure required to counterbalance the elastic recoil of the respiratory system at end-exhalation. Because of time constant inequalities between lung units, the end-expiratory elastic recoil may not be distributed homogeneously.⁹² It follows that PEEPi measured with the counterbalance method represents the pressure required to start inspiratory flow in lung units with a short time constant and fast exhalation (minimum PEEPi). This characteristic of dynamic PEEPi raises several points. First, dynamic PEEPi underestimates the magnitude of static PEEPi, at times by as much as 90%.⁹³ Second, the true clinical impact of PEEPi can be misjudged if only dynamic PEEPi is considered.¹⁵ Third, the difference between dynamic and static PEEPi—often reported as the ratio of dynamic to static PEEPi or “inequality index”¹⁵—can estimate the severity of time constant inequalities of the respiratory system.⁹³ In a study of paralyzed, ventilated patients, Maltais et al⁹³ reported a lower inequality index in patients with airway obstruction, 0.36 ± 0.06 , than in patients without airway obstruction, 0.87 ± 0.05 .

Figure 31-8

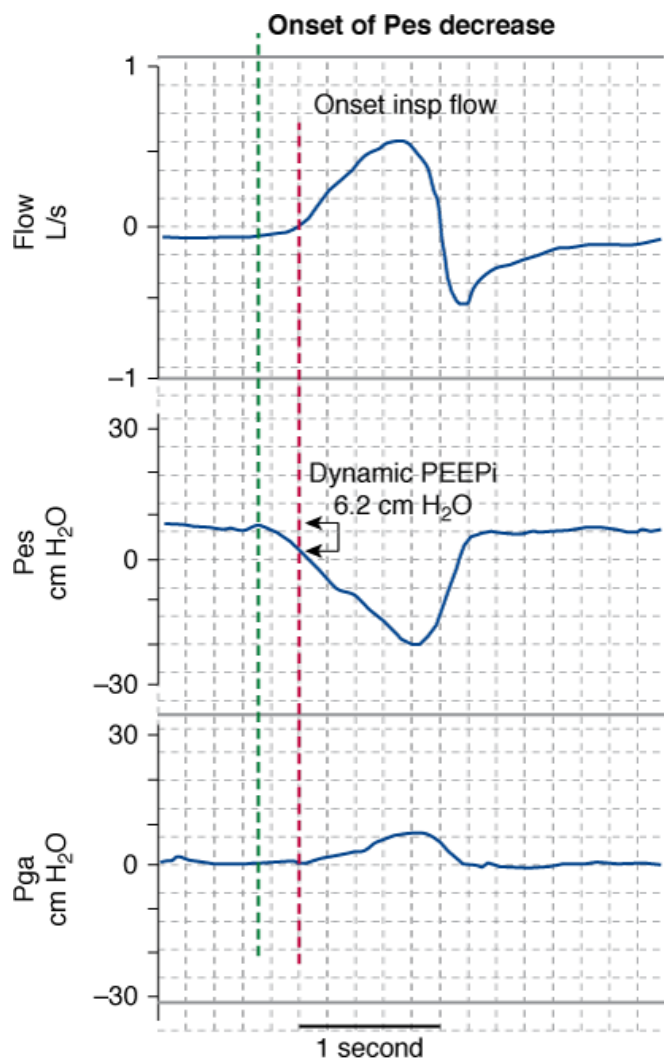


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Inspiratory effort during end-expiratory airway occlusion maneuver. Recordings of flow (*inspiration upward*), airway pressure (*Paw*) and esophageal pressure (*Pes*) in a patient with chronic obstructive pulmonary disease (COPD) in respiratory failure receiving mechanical ventilation. Shortly after the start of airway occlusion (*green vertical line*), the patient exhibits an inspiratory effort (negative deflection in the *Paw* and *Pes* signals). Measurement of static intrinsic positive end-expiratory pressure during active inhalation is inaccurate.

Figure 31-9



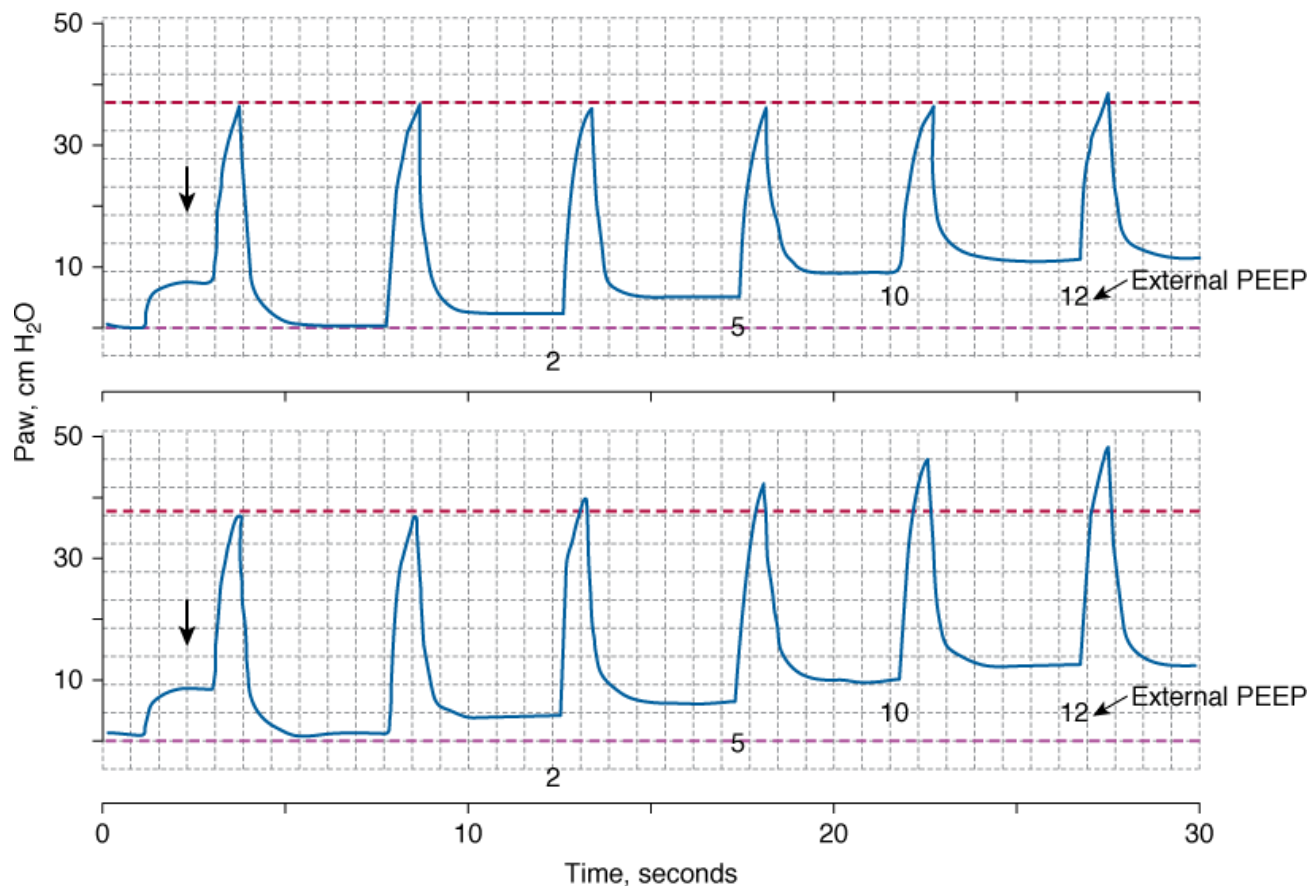
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Recordings of flow (*inspiration upward*), esophageal pressure (*Pes*) and gastric pressure (*Pga*) illustrating the counterbalance method for measuring dynamic intrinsic positive end-expiratory pressure (*PEEPi*) during spontaneous respiration. The *green vertical line* indicates the onset of inspiratory effort and the *purple vertical line* indicates the onset of inspiratory flow. Dynamic *PEEPi* is measured as the negative deflection in *Pes* from the start of inspiratory effort to the onset of inspiratory flow. The *Pga* signal is flat during exhalation suggesting absent expiratory muscle recruitment.

A final assumption with the counterbalance method is that exhalation is passive. If expiratory muscle contraction is present at end-exhalation, the decrease in esophageal pressure at the start of inhalation will reflect, in part, the relaxation of the expiratory muscles rather than inspiratory muscle contraction alone.²⁷ Several techniques to correct for this confounder have been proposed^{74,94} with various success.^{95,96} An additional confounding factor when measuring dynamic *PEEPi* is breath-to-breath variability in the duration of neural *T_E*.⁹⁴ This phenomenon causes a variable duration of lung emptying, which, in turn, can cause large fluctuations of *PEEPi*, even over short periods of time.^{67,94} Systematic investigations of the effect of this additional confounder on the determination of dynamic *PEEPi* are not available.

In summary, the major goal of using external PEEP in patients with airflow obstruction who are triggering the ventilator is to decrease workload¹⁴ and dyspnea⁸⁷ and to improve patient-ventilator synchrony.⁶⁷ Titration of external PEEP should continue until dyspnea has improved, respiratory motor output has decreased,⁹⁷ or until peak and plateau airway pressure demonstrate a sizeable increase (volume-cycled ventilation) (Fig. 31-10)⁹⁸ or tidal volume starts to decrease (PCV).¹⁵

Figure 31-10



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Schematic of airway pressure (P_{aw}) versus time during controlled mechanical ventilation as external positive end-expiratory pressure (PEEP) is increased from 0 to 12 cm H_2O in a patient with expiratory flow limitation (*upper panel*) and in a patient without expiratory flow limitation (*lower panel*). In both patients, static intrinsic PEEP ($PEEP_i$), determined by the end-expiratory airway occlusion technique (↓), is 10 cm H_2O . In the presence of expiratory flow limitation (*upper panel*), peak airway pressure increases only when external PEEP is 12 cm H_2O (dotted red line). In the absence of expiratory flow limitation (such as in patients with asthma; *lower panel*), application of external PEEP causes a proportional increase in peak airway pressure. The qualitative responses to external PEEP, seen with peak airway pressure, are expected to occur also with plateau pressure.

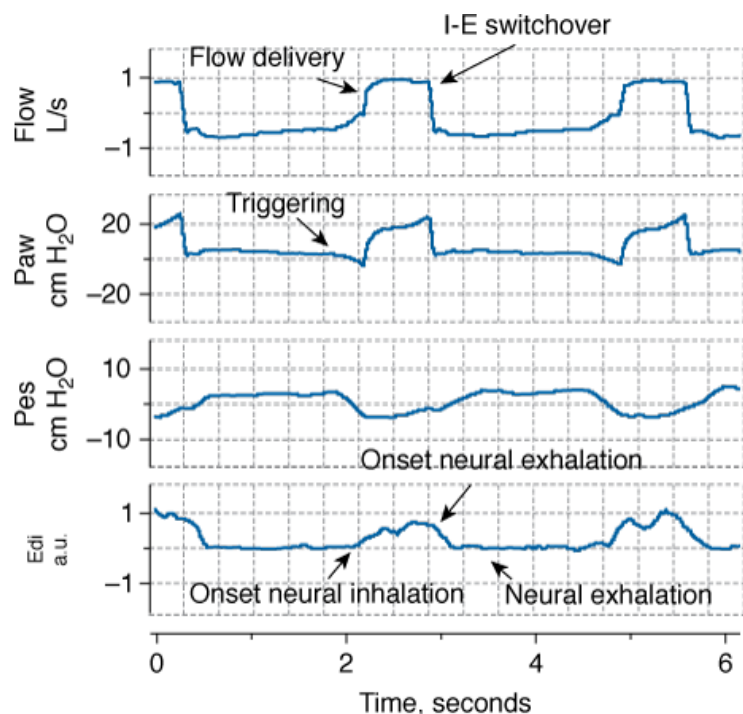
Susceptibility to Ventilator Complications

Patients with COPD are particularly susceptible to ventilator complications. This susceptibility results from several mechanisms, including the pathophysiologic derangements that contribute to the need for mechanical ventilation in COPD (dynamic hyperinflation), side effects of medications, and complications that arise from the underlying lung disease (tissue destruction of the emphysematous lung).

Dyssynchronies

The complex abnormalities of lung mechanics in patients with COPD and respiratory failure make them particularly susceptible to develop patient-ventilator dyssynchrony. Dyssynchronies occur when there is inadequate matching between patient demand and ventilator support in terms of timing, volume, and flow. Dyssynchronies can occur at the onset of neural inhalation (ineffective trigger, trigger delay), during neural inhalation (double triggering, inadequate pressurization, inadequate flow or volume), at the offset of neural inhalation (premature or prolonged assist), and during neural exhalation (autotriggering) (Fig. 31-11).¹⁶

Figure 31-11



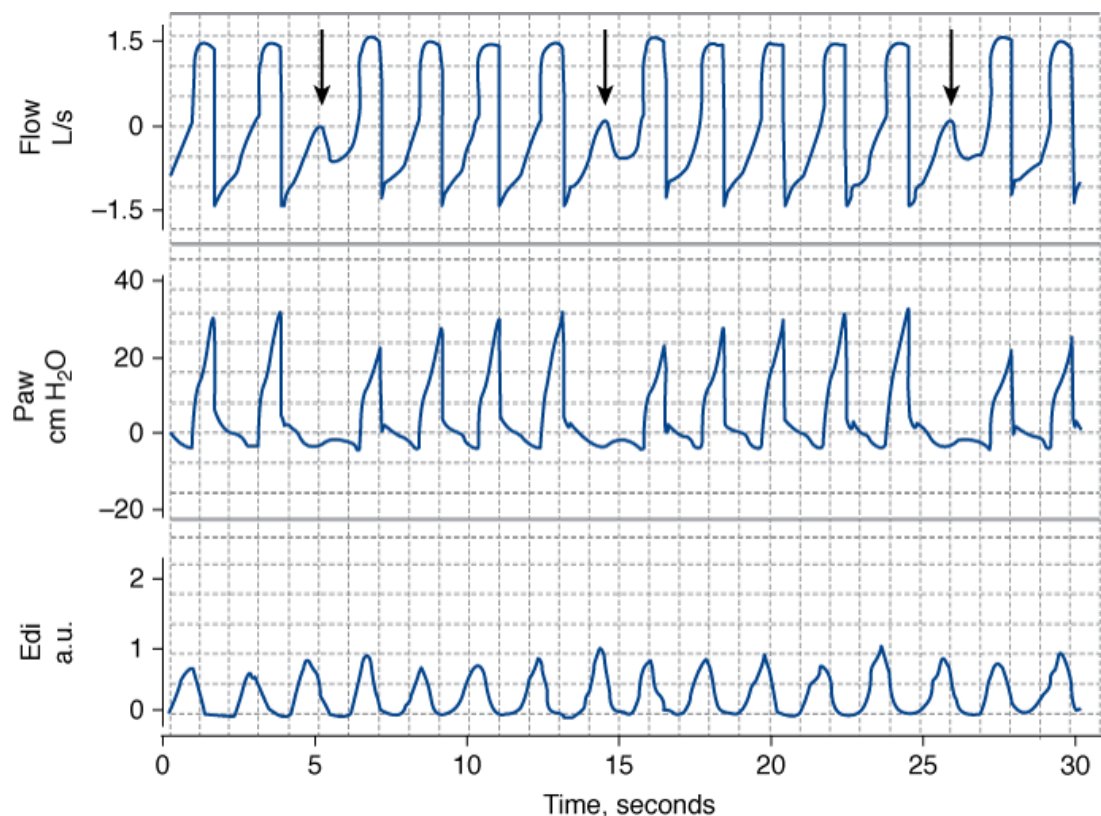
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Points at which patient-ventilator dyssynchrony may arise: onset of neural inhalation (point of triggering, cycling-on function), during neural inhalation (inspiratory flow delivery, posttrigger inflation), onset of neural exhalation (switchover between inspiration and expiration, cycling-off function), and during neural exhalation (autocycling). *a.u.*, arbitrary units; *Edi*, electrical activity of the diaphragm; *Paw*, airway pressure; *Pes*, esophageal pressure.

Ineffective triggering attempts occur in a quarter to a third of inspiratory efforts of patients with COPD receiving high levels of PSV or ACV (Fig. 31-12).⁶⁷ The number of ineffective triggering attempts increases in direct proportion to the level of ventilator assistance and result from premature inspiratory efforts that are insufficient to overcome the increased elastic recoil associated with dynamic hyperinflation.⁶⁷ Because there is no lung inflation during a failed triggering attempt, mechanical exhalation continues for a longer time and end-expiratory volume continues to fall until triggering becomes successful.

Figure 31-12



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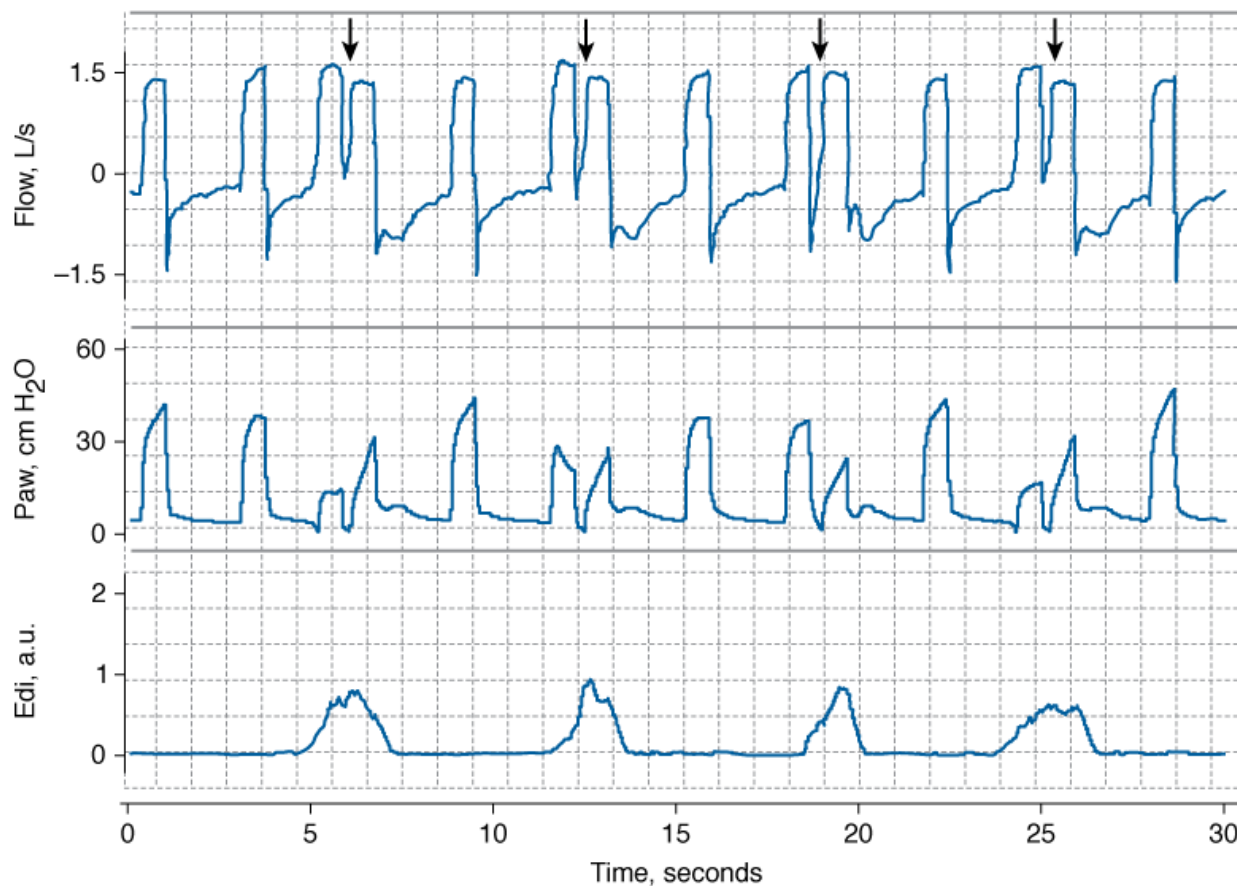
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Failure to trigger the ventilator. Flow, airway pressure (Paw) and electrical activity of the diaphragm (Edi) in a patient with COPD who is receiving assist-control ventilation. Contractions of the inspiratory muscles during the failed triggering attempts ($\downarrow$) cause brief deceleration of expiratory flow and decreases in Paw . The brief decelerations in flow are followed by brief accelerations of expiratory flow that coincide with the termination of the unsuccessful inspiratory effort. Of note, the peak airway pressure increases following each episode of failure to trigger the ventilator. This finding is caused by progressive air trapping. *a.u.*, arbitrary units.

Trigger delays can be caused by factors intrinsic to the ventilator and factors intrinsic to the patient. Ventilator factors include the algorithms that control cycling and the electrical and mechanical processes in the machine.⁹⁹ Patients' factors include respiratory mechanics, respiratory drive, and inspiratory pressure output.^{67,100-102}

A mismatch of the timing between a relatively long neural inhalation and a relatively short mechanical inflation (double triggering) (Fig. 31-13), or as a mismatch between patient flow demand and the flow delivered by the ventilator, can produce dyssynchronies during neural inhalation. Flow mismatch can be either in excess or in defect of the patient's demand. Excessive flow delivered by the ventilator will always raise peak airway pressure. If the patient is receiving NIPPV, an increase in peak airway pressure can cause air leaks around the mask. These leaks are uncomfortable and they diminish the patient's acceptance of NIPPV.⁷⁹ In addition, air leaks can decrease alveolar ventilation and worsen patient-ventilation interaction.

Figure 31-13



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Double triggering caused by long duration of inspiratory effort. Four incidents of double triggering, each indicated by an arrowhead ($\downarrow$). Flow, airway pressure (Paw) and electrical activity of the diaphragm (Edi) in a patient with chronic obstructive pulmonary disease (COPD) and pneumonia who was receiving assist-control ventilation. The patient generates only four neural inhalations (positive deflection in the Edi signal). The duration of each neural inhalation was substantially longer than the duration of mechanical inflation. This mismatch is responsible for double triggering. *a.u.*, arbitrary units.

When the ventilator's inspiratory flow is insufficient for the patient's flow demands, two types of dyssynchrony can occur. First, the ventilator unloads the respiratory muscles unsatisfactorily or it may even impose an external elastic load on the respiratory muscles.^{103,104} Second, the time required to deliver a given V_T will exceed the duration of neural T_I . In this case, mechanical inflation will continue during neural T_E . The sense of being unable to empty the lungs may cause patients to activate the expiratory muscles. This uncomfortable activation will cause a positive spike in the airway pressure signal at end-inhalation.

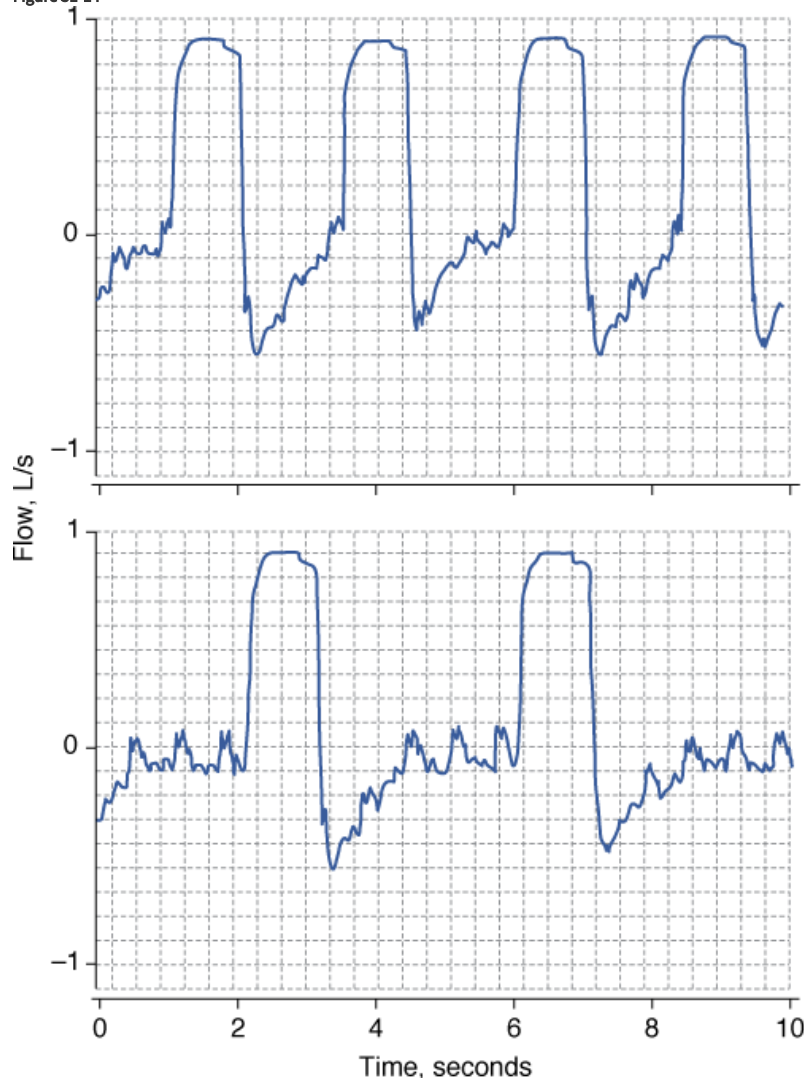
During ACV or PCV, when a patient's neural T_I is short, ventilator inflation may continue into neural exhalation and thus decrease the time available for lung emptying. This phenomenon increases the likelihood for dynamic hyperinflation (with attendant ineffective triggering) and activation of the expiratory muscles.

During PSV, dyssynchronies at the offset of neural inhalation are caused by the algorithm for "cycling-off" of mechanical inflation (see section [Choice of Ventilator Mode](#) above).^{76,77,105} Air leaks between the mask and the patient's face and increased dead space make the synchronization between the offset of neural inhalation and the offset of mechanical inflation particularly tenuous when delivering PSV with NIPPV.^{106,107} Strategies designed to improve synchronization include the use of a time-cycled expiratory trigger¹⁰⁷ and neurally adjusted ventilatory assist.^{101,102,108}

Occasionally a ventilator delivers an assisted breath during neural exhalation. This phenomenon, known as *autotriggering* or *autocycling*, can cause severe respiratory alkalosis.¹⁰⁹ Autotriggering is relatively common when there is water in the ventilator circuit or when air leaks are present.^{99,110} These leaks can occur around the endotracheal tube cuff,¹¹¹ around NIPPV masks,^{99,112} and from chest

tubes.¹⁰⁹ Oscillation of gas in the airways caused by cardiac contractions can also result in autotriggering (Fig. 31-14). Elimination of water from the ventilator circuit,¹¹⁰ reduction of inspiratory trigger sensitivity, and minimization of leaks can reduce autotriggering.^{99,109} An additional cause of autotriggering occurs when a stiff nasogastric tube placed on suction is erroneously passed into the airway rather than into the esophagus.¹¹³

Figure 31-14



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Autotriggering caused by cardiac oscillations. Recording of flow (*inspiration upward*) in a patient with chronic obstructive pulmonary disease (COPD) and aortic insufficiency. The patient was maintained on assist-control ventilation with a rate of 15 breaths/min and flow-triggering. *Upper panel:* The triggering sensitivity was set at 1 L/min. Despite the absence of clinically detectable inspiratory efforts, the ventilator rate was 24 breaths/min. *Lower panel:* When the triggering sensitivity was changed to 4 L/min, the ventilator rate decreased to 15 breaths/min, and large cardiac oscillations could be appreciated on the flow signal.

Ventilator-Induced Hyperinflation

In patients with COPD, mechanical ventilation itself increases the risk of dynamic hyperinflation through several mechanisms. First, ventilator settings (excessive minute ventilation, V_T , short time available for exhalation) can be unsuitable for the respiratory system resistance, elastance, and flow limitation of a given patient.¹⁵ Second, the ventilator's circuit increases flow resistance. This increase in (external) resistance is caused by the combination of resistance of the endotracheal tube (particularly if the tube has concretions or is partially bent),¹¹⁴ ventilator tubing, exhalation valves, and external PEEP.¹⁵ Even in an intubated patient with healthy lungs, total

expiratory resistance (across lungs, endotracheal tube, and exhalation valves) can exceed 10 cm H₂O/L/s (normal: <4 cm H₂O/L/s), enough to cause measurable PEEPi when the minute ventilation exceeds 20 L/min.⁹⁸

Ventilator-Associated Pneumonia

Ventilator-associated pneumonia is a common and potentially lethal complication of invasive ventilation. Factors that directly contribute the development of pneumonia include the presence of the endotracheal tube, contamination of the ventilator circuit and respiratory-therapy equipment with bacteria, ineffective cough, need of sedation, and development of acute sinusitis (see [Chapter 46](#)).

Steroid Myopathy

Most patients with exacerbations of COPD receive corticosteroids.^{115,116} Administration of corticosteroids—with or without concomitant use of muscle relaxants—can trigger development of an acute myopathy characterized by loss of myosin heavy chains within muscle fibers.²⁸ Occasionally, corticosteroids cause acute rhabdomyolysis or myonecrosis.²⁸ In a prospective study of twenty-six patients with COPD exacerbation requiring invasive ventilation, Amaya-Villar et al¹¹⁷ reported development of myopathy in nine patients (35%). Severity of illness (admission Acute Physiology and Chronic Health Evaluation [APACHE] II score), occurrence of sepsis and total dose of corticosteroids were higher in patients who develop myopathy than in those who did not develop it. Myopathy was associated with an increase in the duration of ventilation and lengths of ICU and hospital stay. An association between corticosteroid administration and development of myopathy in the ICU¹¹⁸ or between myopathy and duration of mechanical ventilation¹¹⁹ has been reported by other investigators.

In patients with steroid myopathy, acute weakness is likely caused by impaired muscle membrane excitability.¹²⁰ In contrast, prolonged weakness is more likely caused by proteolysis of myosin filaments, reduced myosin transcription, and decreased expression of the sarcoendoplasmic reticulum calcium-adenosine triphosphatase (SERCA)-type pumps (key protein pumps for calcium kinetics during muscle relaxation following contraction, which scavenge calcium from the cytosol).²⁸

Two recent observations raise questions on the negative impact of (short-term) administration of corticosteroids during mechanical ventilation. In a study conducted in rats, Maes et al¹²¹ reported that a single high dose of methylprednisolone protected diaphragmatic muscle function from the deleterious effects of controlled mechanical ventilation. According to the investigators, the most likely mechanism for this protective effect was a corticosteroid-dependent inhibition of the calpain proteolytic system. The second investigation was conducted in 340 patients with severe acute respiratory distress syndrome.¹²² Patients were randomly assigned to receive, for 48 hours, either the neuromuscular blocking agent cisatracurium besylate (178 patients) or placebo (162 patients); 42% of all patients received corticosteroids for sepsis. Early administration of cisatracurium besylate—with or without corticosteroids—improved adjusted 90-day survival and increased the time off the ventilator without increasing muscle weakness.

Weaning from Noninvasive Positive-Pressure Ventilation

Most investigators decrease NIPPV when respiratory distress and tachypnea have resolved and FiO₂ requirement is less than 0.40.⁵⁵ At that point, inspiratory and expiratory pressures are decreased by 2 to 3 cm H₂O each hour as tolerated. When gas exchange is adequate and patients are comfortable with inspiratory pressure of 10 to 12 cm H₂O and external PEEP of 3 to 5 cm H₂O, clinicians can temporarily discontinue NIPPV. Periods of unassisted breathing are maintained until patients develop some degree of respiratory deterioration (tachypnea, dyspnea, increased use of accessory muscles).¹²³ Ideally, NIPPV is interrupted for progressively longer periods until patients can sustain unassisted breathing indefinitely.^{51,55,123} In patients with exacerbations of COPD with an initial pH of 7.13 to 7.29 and who respond successfully to NIPPV, the mean duration of ventilator assistance—plus periods of interruption—can range from 2 to 9 days.^{55,57,59,123,124} Occasionally patients require nocturnal noninvasive ventilation.

Weaning from Invasive Ventilation

Chapter 58 discusses ventilator weaning. In brief, my personal approach is to screen for early weanability with use of the ratio of respiratory frequency to tidal volume (f/V_T).^{125,126} When the result predicts weaning success, I proceed to a T-tube trial, usually lasting no more than 30 minutes. If a patient tolerates this trial without undue distress, I proceed to extubation.

Adjunctive Therapies

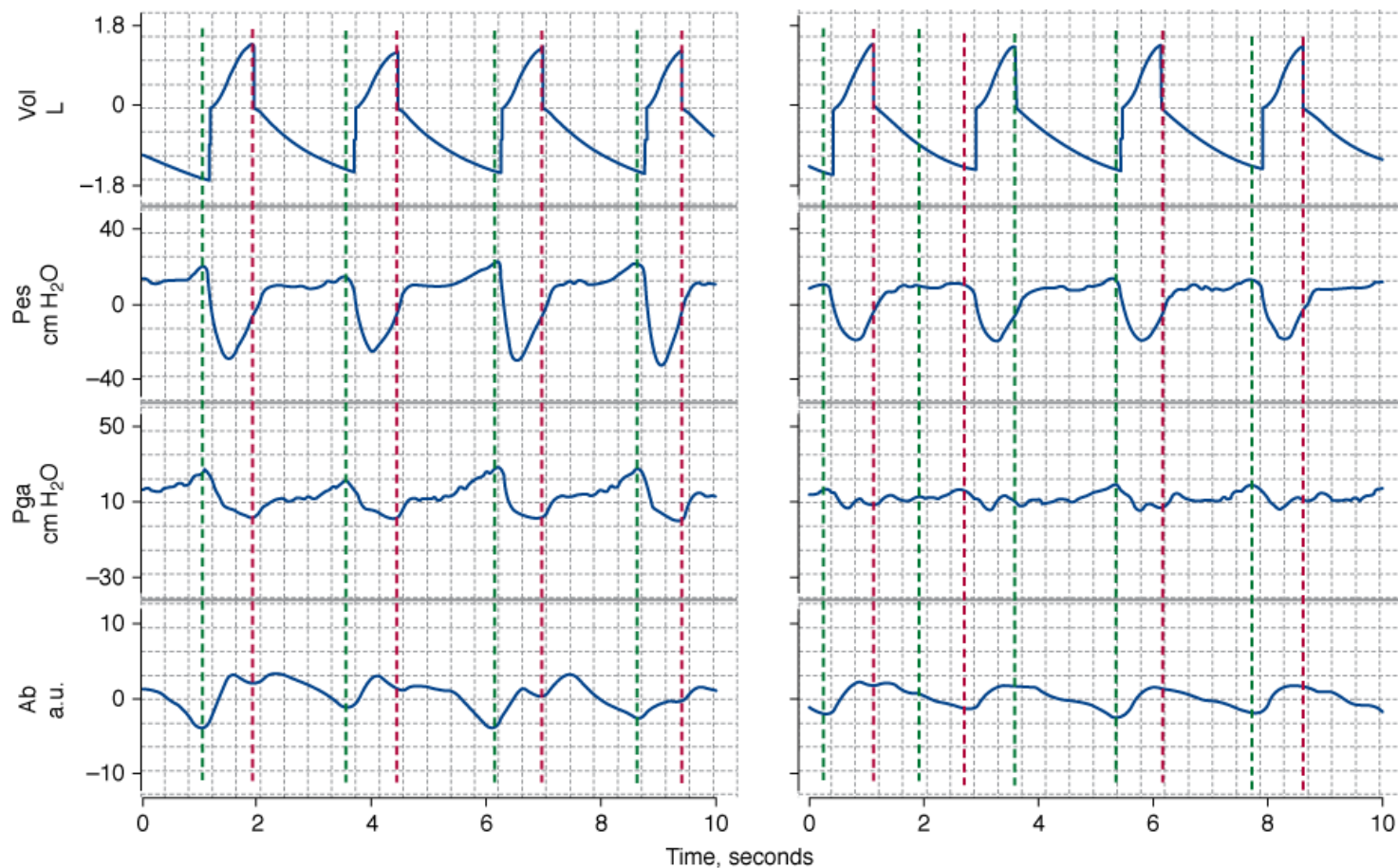
Several adjunctive therapies are administered in ventilated patients with COPD. These include therapies aimed at improving airflow (bronchodilators, helium–oxygen mixtures), decreasing inflammation (corticosteroids), and controlling infections (antibiotics).

Improvement in Airflow: Helium–Oxygen

Gas mixtures of helium and oxygen have lower density and slightly higher viscosity than standard air–oxygen mixtures.¹²⁷ This lower density decreases the Reynolds number of helium–oxygen mixtures.¹²⁷ A smaller Reynolds number can decrease resistance to gas flow through several mechanisms.¹²⁷ First, it will make flow laminar over a larger number of (medium-to-large) airway generations, over a greater fraction of the respiratory cycle, and over higher flow rates.¹²⁷ Second, when gas enters a bifurcated airway, laminar flow results in a lower airway resistance than does turbulent flow.¹²⁷ Third, for a given rate of laminar flow, a lighter gas mixture (smaller Reynolds number) results in a shorter entrance length. Entrance length is the length of a tube (bronchial tube) required to reestablish laminar flow after gas enters a bifurcated airway. For the portion of a tube over which laminar flow is becoming established, resistance will be higher as compared with when laminar flow has already been established.¹²⁷

Several investigators have assessed the possibility to use those mixtures in patients with COPD.^{44,51,128–132} In stable patients with severe COPD, Hussain et al¹²⁸ reported that patients who exercised while breathing helium–oxygen experienced less inspiratory muscle load, dyspnea, and dynamic hyperinflation, and could exercise for longer than when breathing air–oxygen alone (Fig. 31-15).¹²⁸ In patients with COPD and respiratory failure, NIPPV plus helium–oxygen decreases work of breathing and dyspnea and improves CO₂ clearance more than NIPPV alone.^{129,130}

Figure 31-15



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Helium-oxygen decreases inspiratory effort during exercise. Tracings of volume, esophageal (*Pes*) and gastric (*Pga*) pressures, and cross-sectional area of the abdomen (*Ab*) in a patient with severe chronic obstructive pulmonary disease (COPD) and severe hyperinflation during submaximal constant-load exercise test (cycle ergometer). The *green vertical lines* indicate the onset of inspiratory effort and the *red vertical lines* indicate the end of tidal inspiration. The patient exercised while breathing 30% *oxygen* (*left panel*) or a mixture of 70% helium plus 30% *oxygen* (*right panel*). Tidal volume was similar in the two conditions but tidal swings in *Pes* were decreased by almost half during helium-oxygen as contrasted with *oxygen* alone. In addition, expiratory muscle recruitment, reflected by positive deflection in the *Pga* signal and negative deflection in the *Ab* signal during exhalation, was less with helium-oxygen (*right panel*) than with *oxygen* alone (*left panel*). *a.u.*, arbitrary units. (Data from Hussain et al.¹²⁸)

Two prospective trials have been conducted to assess whether NIPPV plus helium-oxygen could reduce the intubation rate and improve the clinical outcome of patients with COPD in respiratory failure.^{51,131} In 123 patients, Jolliet et al¹³¹ used a mixture of 78% helium and 22% *oxygen*. Intubation rates and ICU length of stay were comparable. In 204 patients, Maggiore et al⁵¹ used a 65% helium and 35% *oxygen* mixture and found no differences in arterial blood gases, dyspnea, duration of NIPPV, ICU and hospital length of stay, and 28-day mortality.

At least two mechanisms may have contributed to these negative results.^{51,131} First, the trials were underpowered because of lower-than-expected intubation rates in the control groups. Second, helium-oxygen can improve gas dynamics by promoting the transition of flow from turbulent to laminar and by decreasing the driving pressure in areas where flow remains turbulent.¹³² Turbulent flow is more likely when the increased resistance to airflow is located in the central airways than in the peripheral airways. If the increased resistance was mainly located in the peripheral airways of the patients in these two trials,^{51,131} it would have been impossible for helium-oxygen to improve clinical outcomes.

Improvement in Airflow: Bronchodilators

[Chapter 63](#) discusses bronchodilator therapy. My personal approach in patients receiving either NIPPV or invasive ventilation is to administer short-acting bronchodilators (albuterol and [ipratropium](#)) with a metered-dose inhaler and spacer. Occasionally I use nebulizers in patients receiving NIPPV.¹³³

Corticosteroids

Administration of systemic corticosteroids in patients with exacerbations of COPD reduces dyspnea, treatment failure, and hospital stay.¹¹⁵ Systemic corticosteroids are also associated with improvement of forced expiratory volume in 1 second and arterial blood-gases during the early treatment phase (to 72 hours) and at the end of treatment (up to 15 days).¹¹⁵ Systemic corticosteroids, however, predispose to hyperglycemia and do not decrease mortality.¹¹⁵ Dose and duration of corticosteroid therapy continues to be debated.¹¹⁵ In this regard, Lindenauer et al¹¹⁶ compared the outcomes of nearly 80,000 exacerbations of COPD treated with low doses of corticosteroids administered orally (20 to 80 mg of prednisone) versus higher doses administered intravenously (120 to 800 mg prednisone equivalent). The risk of treatment failure, inpatient mortality, readmission for acute exacerbation of COPD was equivalent in the two groups. No benefit is derived by prolonging therapy beyond 2 weeks.¹³⁴ In patients with exacerbations of COPD, high-dose nebulized [budesonide](#) (1.5 to 2 mg every 6 hours) can be as effective as systemic corticosteroids, although much more expensive.¹³⁵

Antibiotics

The use of antibiotics in the treatment of acute exacerbation of COPD is largely accepted.¹³⁶ In patients without pneumonia, inexpensive broad-spectrum antibiotics, such as ampicillin, doxycycline, and trimethoprim-sulfamethoxazole, are usually effective.¹³⁴ In a randomized, double-blind trial of 170 patients with an acute exacerbation of COPD requiring mechanical ventilation, trimethoprim-sulfamethoxazole was equivalent to a more expensive agent (ciprofloxacin).¹³⁶ In a recent review of eleven trials involving 917 patients, antibiotic administration in exacerbations of COPD was shown to reduce short-term mortality by 77% and the risk of treatment failure by 53%.¹³⁷

Patients with COPD who have community-acquired pneumonia and who do not require ICU admission should be treated with a respiratory fluoroquinolone alone ([levofloxacin](#), [gemifloxacin](#), [moxifloxacin](#)) or with a β -lactam plus a macrolide.¹³⁸ Macrolides are attractive because they possess additional antiinflammatory and mucoregulatory properties.¹³⁹ Patients with COPD and pneumonia who require ICU admission should be treated with a β -lactam (cefotaxime, ceftriaxone, ampicillin-sulbactam) plus [azithromycin](#) or with a β -lactam plus a respiratory fluoroquinolone.¹³⁸

Patients with recurrent exacerbations leading to frequent use of steroids and antibiotics are at increased risk of *Pseudomonas* pneumonia.¹³⁸ When these patients develop pneumonia, they should be treated with a β -lactam with antipseudomonas and antipneumococcal properties (piperacillin-tazobactam, cefepime, imipenem or meropenem) plus [ciprofloxacin](#) or [levofloxacin](#). An aminoglycoside plus [azithromycin](#) can be used as a alternative to [ciprofloxacin](#) or [levofloxacin](#).¹³⁸ When community acquired methicillin-resistant *Staphylococcus aureus* is a consideration, vancomycin or linezolid should also be administered.¹³⁸

Other Adjunctive Therapies

In patients with COPD and respiratory failure, mucolytics agents, respiratory stimulants, and methylxanthines are of uncertain benefit or no benefit or even harmful.^{140,141} Chest physiotherapy is ineffective or even detrimental in patients with exacerbations of COPD requiring medical treatment alone or medical treatment plus invasive ventilation.¹⁴¹ In patients treated with NIPPV, intrapulmonary percussive ventilation⁵⁶ and exhalation under positive-pressure (PEP mask)¹⁴² may improve gas exchange, enhance removal of airway secretions and reduce duration of ventilator support and ICU length of stay.

Patients with COPD and acute respiratory failure often experience anxiety and agitation.¹¹⁰ These symptoms can be caused by many factors including dyspnea, hypoxia, infection, fever, fear, and sleep deprivation. Identification and treatment of the underlying cause of anxiety does not always result in a satisfactory control of these symptoms. Anxiety and agitation can be sufficiently severe to interfere

with the patient's adaptation to noninvasive or invasive ventilation. In these patients, the physician should consider the judicious use of sedatives.¹¹⁰ The use of sedatives during NIPPV, however, is potentially dangerous and thus controversial.¹¹⁰

Long-Term Outcome Following Acute Respiratory Failure Treated with Mechanical Ventilation in Chronic Obstructive Pulmonary Disease

Patients with COPD who survive an episode of acute respiratory failure treated with NIPPV have significant morbidity and mortality following hospital discharge.^{124,143} Chu et al¹²⁴ undertook a prospective study to assess the long-term outcome of 110 patients with COPD who survived an episode of acute respiratory failure treated with NIPPV. One year later, nearly half of the patients were dead, mostly because of respiratory failure. It is uncertain whether domiciliary nighttime NIPPV could improve the long-term outcome of these patients.¹⁴⁴

Long-term use of nocturnal NIPPV has also been proposed for stable patients with severe COPD and chronic ventilatory failure,^{145–147} including those with recurrent admissions for acute respiratory failure.¹⁴⁸ In two Italian studies, long-term nocturnal NIPPV plus oxygen was associated with small improvements in dyspnea,^{146,147} daytime hypercapnia, and health-related quality of life.¹⁴⁷ More recently, the investigators of the Australian trial of noninvasive Ventilation in Chronic Airflow Limitation (AVCAL)¹⁴⁵ concluded that nocturnal NIPPV in stable oxygen-dependent patients with hypercapnic COPD may improve survival while worsening quality of life (see [Chapters 18](#) and [68](#)).

In a recent multicenter study conducted in France and Tunisia, Girault et al¹⁴⁹ reported the outcome of more than 200 patients with COPD who had been intubated for an episode of acute respiratory failure. In the study, 76% of patients survived hospitalization. At discharge, 81% of the survivors were able to maintain spontaneous respiration completely unassisted, 12% were discharged on noninvasive ventilation, and 4% required invasive ventilation through a tracheostomy.

Tracheostomized patients discharged from an ICU may be transferred to a long-term weaning facility.^{150,151} In approximately one-third of patients transferred to these facilities, mechanical ventilation can be successfully discontinued after early trials of spontaneous respiration.¹⁵¹ In the remaining patients, the weaning process is slow (average of 33 to 44 days).^{150,151} Eventually, 55% to 75% of patients in whom the weaning process is slow are successfully weaned and decannulated.^{150,151} The remaining 18% to 45%, however, die during the admission to the weaning facility or remain ventilator dependent.^{150,151}

The long-term outcome of patients with COPD who are ventilator dependent is poor.¹⁵⁰ In a small series of patients with COPD who required prolonged mechanical ventilation, Nava et al¹⁵⁰ reported death rates of 50% at 6 months and 78% at 2 years in patients who could not be weaned. In contrast, death rates in patients who were weaned were 20% at 6 months and 32% at 2 years ($p < 0.01$).

Conclusion

Few groups of patients have benefitted more from the intelligent use of mechanical ventilation than have patients with COPD. Forty years ago, pulmonologists commonly employed little more than palliative care in the management of patients with COPD who had developed acute respiratory failure. Pulmonologists were reluctant to intubate such patients because of their increased susceptibility to complications and the fear that subsequent extubation would be well-nigh impossible. All that has changed. Expert use of mechanical ventilation helps the vast majority of patients with COPD to successfully battle an episode of acute respiratory failure, and patients are typically weaned and extubated without undue difficulty. Indeed, ventilator assistance is mostly achieved noninvasively, without recourse to intubation. Success with use of NIPPV has not been achieved to the same extent with any other group of patients.

The successes achieved by mechanical ventilation are the consequences of major advances in understanding of respiratory pathophysiology that began during World War II and continued in the subsequent decades. Harnessing this knowledge to better engineering of ventilator machinery combined with more sophisticated methods of patient monitoring has resulted in more intelligent use of mechanical ventilation and fewer complications. Selecting the most appropriate mode of ventilator assistance and settings in a given patient with COPD is predicated on careful evaluation of that patient's pathophysiology. Few other illnesses managed in an ICU

depend on the same level of physiologic understanding as does the management of a patient with COPD in acute respiratory failure—the reason why I have devoted the bulk of this chapter to discussion of the pathophysiology of COPD.

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